BUILDING THE TWIN

Dynamic inputs

Module 4 · The senses of a twin

Point or network?

Point sensors

Loops and radar

Cameras at junctions

Accurate, local

Network sources

Probe GPS speeds

Bus locations

Wide, but sampled

Every source has a blind spot

- Loops: one point, need road works
- Cameras: light, weather, occlusion, privacy
- Probe data: sample bias, purchased
- V2X: few equipped vehicles today

Four dimensions of data quality

- Completeness: is anything missing?
- Accuracy: is it right?
- Timeliness: is it fresh enough?
- Consistency: do sources agree?

Standards for live data

- GTFS-Realtime and SIRI: public transport
- DATEX II: road traffic in Europe
- SAE J2735: vehicle-to-infrastructure messages
- NGSI-LD: smart city context data

Fusion

1 + 1 > 2

Combine sources so each covers another's blind spot